The Estimator Decides:

What Curricula and Failure Recycling Can — and Cannot — Do for RL with Verifiable Rewards

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Code, logs, and proofs: https://github.com/linjiw/curriculum-maxrl

Abstract

Post-training with verifiable rewards increasingly relies on two data-level interventions: *difficulty curricula*, which reallocate rollout budget toward learnable prompts, and *failure recycling* (hindsight relabeling), which converts failed rollouts into verified successes of the tasks they actually achieved. Both are typically treated as objective-agnostic add-ons. We show they are not — **the advantage estimator underneath decides whether each helps or harms**, and its algebra says so in closed form. For success-conditioned (MaxRL-style) advantages, the expected advantage mass a prompt emits from N rollouts is exactly $2(\text{pass@}N - \text{pass@}1)$: a compute-indexed zone-of-proximal-development functional that derives the curriculum's band, width, and scaling from the rollout budget alone (published learnability curricula are its N=2 slice); RLOO's mass is $2p(1-p)$; GRPO's per-question weight decays only as $\sqrt{p(1-p)}$, overweighting both tails. This predicts an estimator-conditioned divergence we then measure: across a 1.26M-parameter maze suite spanning four teacher conditions, every MaxRL run grows pass@k coverage and every GRPO run loses it — an estimator main effect on coverage (permutation p=0.0079, nine runs, zero exceptions). A pre-registered LLM-scale 2×2 on GSM8K landed the predicted sign on its registered run — GRPO+teacher the only regressing cell — while a replication seed with weaker measured steering did not reproduce it: the interaction at LLM scale is treatment-intensity-dependent and not yet established, a status we report as measured. Recycling, we find, has its own failure mode, previously unreported: on Countdown arithmetic with an exact-verifier relabeler running in production, hindsight arms improve mean@16 while *losing* pass@16 coverage — **recycling-induced sharpening**, the data-induced sibling of GRPO's weight-induced collapse: verifier-filtered self-imitation concentrates the policy on reachable outputs and spends exploration breadth. The same algebra supplies the mitigation: a relabel to a destination the model already reaches sits at $p \to 1$, where the mass functional is zero, so we gate relabel admission by the destination's estimated pass rate. Over three seeds the gate restores frontier-tier pass@16 to baseline while keeping ∼60% of recycling's mean gain, and sweeping its strength traces a monotone mean-vs-coverage frontier — a dial, not a switch. One diagnosis covers both results: audit your estimator before you ship a curriculum — or a recycler — and measure coverage, the currency in which both failure modes are visible.

1 Introduction

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, generation dominates wall-clock (87% of step time, measured across our GSM8K runs), and on hard task pools most of that compute buys nothing — under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal**: either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field's response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets [8], target-difficulty controllers [9], category bandits [10], learnability-based rejection sampling [11], dead-prompt filtering [12, 14]. The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently brought into RLVR-style loops for robot policies [18]. Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*:

bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator's own algebra. Three questions organize our results:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected advantage mass a task emits from a group of N rollouts has a closed form — 2(pass@N − pass@1), a compute-indexed zone-of-proximal-development functional (§3). Mass is a *surrogate* for learning signal — it is the estimator-intrinsic, model-free part of the gradient, deliberately agnostic to score-vector geometry (Remark 1) — but it is the part a sampler can act on before spending a rollout. It makes the allocation *derivable* rather than heuristic (the band's location, width, and N-scaling come from the budget you already chose; published learnability curricula are the N=2 slice), and it bounds the channel: allocation can only redistribute signal that exists, a perfect-information allocator ties our cheap posterior in end performance (§7.1), and at LLM scale a prompt-level teacher starves before it pays (§7).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N-1)\times$ more mass than RLOO on frontier tasks as $p \to 0$; GRPO's variance-normalized weights flatten the profile and up-weight both tails. We predict and measure the resulting divergence at two scales: on the maze, every MaxRL run grows pass@8 coverage and every GRPO run loses it, across four teacher conditions (estimator main effect, permutation p=0.0079, nine runs, zero exceptions); on GSM8K, the pre-registered prediction that GRPO+teacher would be the only regressing cell landed on the registered run, and a replication seed with weaker measured steering did not reproduce it (§7.5 reports both). And recycling has its own, previously unreported failure mode: **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@k coverage (§7), the data-induced sibling of GRPO's weight-induced collapse, formally the composition of hindsight's marginal-distribution shift [16] with curation-driven self-consumption [17].

Q3: Does the algebra also supply the fix? Yes: a relabel to a destination the model already reaches sits at $p \to 1$, where the mass functional is zero — recycled updates there buy sharpening, not signal. So we gate relabel admission by the destination's estimated pass rate (the same utility that schedules sampling, applied to the recycler). Over three seeds, ungated recycling lifts tier-1 mean@16 by +.046 while losing .049 of pass@16 (both outside seed noise). The gate's strength interpolates between that trade and recycling-off, and where the useful middle lands is tier-dependent: at the frontier tier, the moderate gate restores coverage to baseline (.279±.019 vs. no-recycling .274) while keeping ∼60% of recycling's mean gain (+.016 of +.026); at the nearer-saturation tier 1 the same setting neutralizes recycling almost entirely, and a full-strength rerun converges to recycling-off behavior on both axes — the mitigation is a dial whose useful range sits where relabel destinations are not yet saturated. The mechanism is visible in-run: the gate's rejection rate rises monotonically (.12 → .85) as the model's reachable set expands into it, admission shifts from novel destinations (73% early) to still-rare revisits (94% late), and the gated arm preserves *more* policy entropy than the no-recycling baseline at 4.9× less relabel dose. Where recycling is healthy the gate is transparent: rerunning the validated CPU testbeds with the gate enabled reproduces their results to within noise (skill-chain AUC .8879 vs. .8876; gridworld bit-identical) because hard-destination relabels pass the gate untouched. One method, no per-domain switches.

Taking that sentence literally produces this paper. Our contributions:

1. **The curriculum band is a theorem, not a heuristic** (§3): the expected advantage mass of a prompt is exactly 2(pass@N − pass@1), a derived ZPD functional whose band location, width, and compute-scaling are set by N alone. Learnability curricula are its N=2 slice; mass-optimal rollout allocation is water-filling on $p(1-p)^N$. We are explicit about the surrogate's scope (Remark 1) — and a correction to the MaxRL paper: the practical drop-K=0 estimator is unbiased for the T=N−1 objective, not T=N.

2. **A practical teacher** (§4): Thompson sampling on a decayed Beta posterior with the derived utility; the per-group advantage estimator is untouched, so each sampled group's gradient remains an unbiased MaxRL gradient for its own prompt — reweighting which prompts are visited changes the training distribution, a shift we state and test rather than hide (Remark 2).

3. **Failure recycling with a verified-success contract and a characterized gradient** (§5): relabeling dead groups to achieved sub-goals yields the ML gradient of the relabeled task under the original sampling's conditional law — exact when the laws match (where we validate, they do: cosine 0.956 vs. 0.958 for

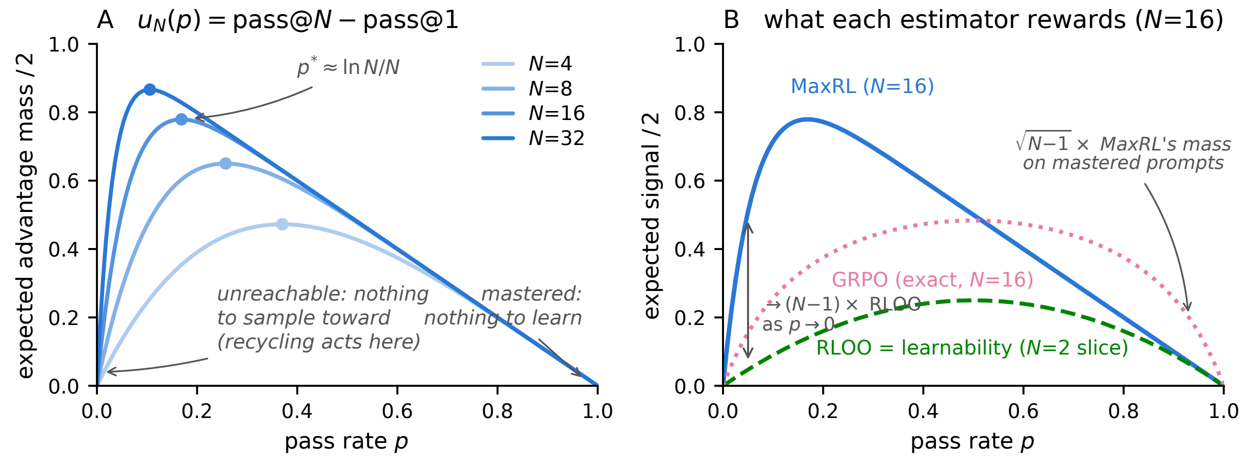

Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1-(1-p)^N) - p$ — the estimator's expected advantage mass (Prop. 1) — for growing rollout budgets N. The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; both tails are dead zones the weight function cannot reach. *Right:* all three curves are exact expected mass at N=16 (GRPO's is $\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$, $K \sim \mathrm{Bin}(N,p)$; MC-verified). MaxRL concentrates $\to (N-1)\times$ more mass than RLOO on frontier prompts (Prop. 2) and releases mastered prompts fastest; GRPO's variance normalization keeps $\sim \sqrt{N-1}\times$ MaxRL's relative mass on mastered prompts — the maintenance-of-easy-tasks its curriculum arms lose in §7. RLOO's profile is exactly the published "learnability" objective — the N=2 slice.

fresh groups), biased otherwise, with two contracts whose violation we price and a dose-vs-direction decomposition of the gain.

4. **A three-channel account with a safety rule** (§6–7): allocation saturates (a matched true-pass-rate oracle ties the cheap posterior); creation is the channel that still adds on top of the oracle; the objective underneath decides whether either is safe — confirmed at LLM scale on a pre-registered prediction.
5. **An honest evidence discipline**: pre-registered predictions, documented negatives with diagnoses, one retraction, and a two-round adversarial audit tracing every number to a log or proof.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1-p)$ and truncates at order T, giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE (T=1) to exact ML ($T \to \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped, which is unbiased for the $T = N-1$ objective. Its weight function grows as $p \to 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 The estimator is the curriculum

Proposition 1 (Advantage mass; MC-verified)**.** *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights,*

$$\mathbb{E}\Big[\textstyle\sum_i |w_i|\Big] \;=\; 2\big(\textit{pass@N}(p) - \textit{pass@1}(p)\big) \;=\; 2\big((1-(1-p)^N) - p\big).$$

Interpretation. *The estimator's expected learning signal on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. That sentence is a curriculum: zero on mastered prompts, zero on* *unreachable ones, maximal at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ — a zone of proximal development whose center, width, and compute-scaling are set by the one parameter you already chose.*

Proposition 2 (Frontier amplification)**.** *As $p \to 0$, the ratio of MaxRL's expected signal to RLOO's ($2p(1-p)$ exactly) tends to $N-1$.*

Interpretation. *The comparison to GRPO is two-sided and exact: GRPO's expected mass is $\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$ with $K \sim \mathrm{Bin}(N,p)$, and taking tails, MaxRL/GRPO $\to \sqrt{N-1}$ as $p \to 0$ while GRPO/MaxRL $\to \sqrt{N-1}$ as $p \to 1$. MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sqrt{N-1}\times$ more of its mass on mastered prompts — maintenance a curriculum then removes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §7 shows the mastered-tail difference becoming an active failure under a curriculum.*

Proposition 3 (Allocation)**.** *The rollout allocation maximizing total expected advantage mass is greedy water-filling on the marginal $p(1-p)^N$ — the probability that the next rollout is a group's first success.*

Interpretation. *This is the mass-optimal* reference *allocator, not the deployed sampler: the practical teacher (§4) keeps fixed group sizes and samples prompts $\propto u^\gamma$, trading optimality for group-parallel batching and for the exploration a point-optimal allocator forfeits (the unconstrained maximizer concentrates on the current arg-max, a brittle choice under posterior noise). We measure what the gap costs: a true-pass-rate water-filling oracle ties the Thompson teacher in end performance (§7.1).*

Remark 1 (Scope of the surrogate). Advantage mass $\mathbb{E}\sum_i |w_i|$ is the model-free factor of the policy gradient $\sum_i w_i \nabla \log \pi(z_i)$: it counts weighted rollouts, not the score-vector geometry (norms, alignment, cancellation) that also shapes the realized update. We use it as the sampling utility for three reasons. (i) It is what a sampler can know *before* spending rollouts — score geometry is only observable after generation, when the budget is already spent. (ii) Its zeros are exact: at $p \in \{0, 1\}$ the estimator emits no gradient of any norm, so the dead zones it predicts are real regardless of geometry. (iii) Empirically its band transfers: the teacher built on it produces the coverage and efficiency effects of §7. What it does not claim is proportionality to expected improvement — that depends on the model, and §7.4a shows a regime (all tasks learnable at budget) where reallocating mass buys nothing.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x\sim q_t}[J(x)]$: each group's gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ. This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t{=}\rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §7 are on fixed held-out pools under ρ, so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ-measured outcome.

One identity, three literatures: learnability curricula [11, 19] are the $N{=}2$ slice of u_N; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling [15] is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). Closest prior: DisCO [2] derives the per-question weights of the GRPO family ($\omega(q) = \sqrt{p(1-p)}$ for GRPO, $p(1-p)$ for Dr. GRPO) to diagnose difficulty bias, and SC-SDPO [3] notes the residual $\sqrt{p(1-p)}$ acts as an implicit curriculum. Our lemma generalizes this line to success-conditioned estimators (where the functional is $u_N = \text{pass@}N - \text{pass@}1$, compute-indexed by N), and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

4 FrontierMax: the schedule

A decayed Beta posterior (decay 0.7) tracks each prompt's pass rate from observed group outcomes only — never from relabeled successes (violating this inflates the posterior: $\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a 10% uniform floor; $\gamma \approx 4$ when tasks share skills (learning compounds — validated on chains, and its non-transfer to broad pools was predicted in advance by a compounding ODE model), $\gamma = 1$ otherwise. Live groups train with unmodified MaxRL advantages. Honest knob inventory: decay, floor, and γ exist, with validated defaults; what is *derived* rather than tuned

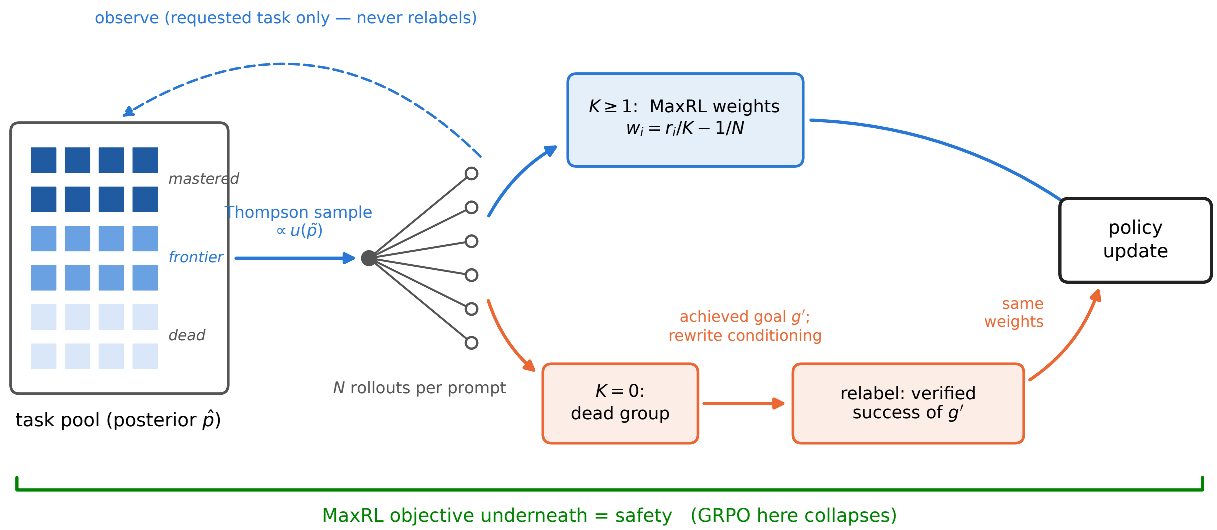

Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

is the difficulty band itself — location, width, and N-scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Proposition 4 (Hindsight update, characterized)**.** *Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields the ML gradient of task g' under the conditional law induced by the* original *task's sampling; it equals the fresh-rollout gradient exactly when the two conditional laws match, and is otherwise a distribution-shifted (biased) estimate of it.*

Interpretation. *So "exactness" is a property to* measure per environment*, not a blanket guarantee. Where we validated it, the laws match by construction (prompt-independent skill execution) and the measurement agrees: per-group cosine to the true relabeled-task gradient 0.956, vs. 0.958 for fresh unbiased groups. In Countdown the relabel keeps the trajectory's own achieved value — the shift enters only through which expressions get attempted for which targets — and the verified-success guarantee (contract 1) holds regardless. What the shift costs in general is an open, environment-specific question.*

Two contracts keep practice on the proof's side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment's own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Recycling is the only mechanism here that *creates* signal, and we decompose its headline gain rather than let it inflate: of the +0.22 AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries +0.037 beyond the strongest placebo and is exactness-dependent (random-target relabels are harmful). Both numbers ship together. The value is regime-dependent in a way we can state precisely: the gain is proportional to how much a relabeled skill can compound — +0.22 AUC on fixed task pools, +0.01 on one-shot task streams. Task-set fixedness is the single most important regime variable we found.

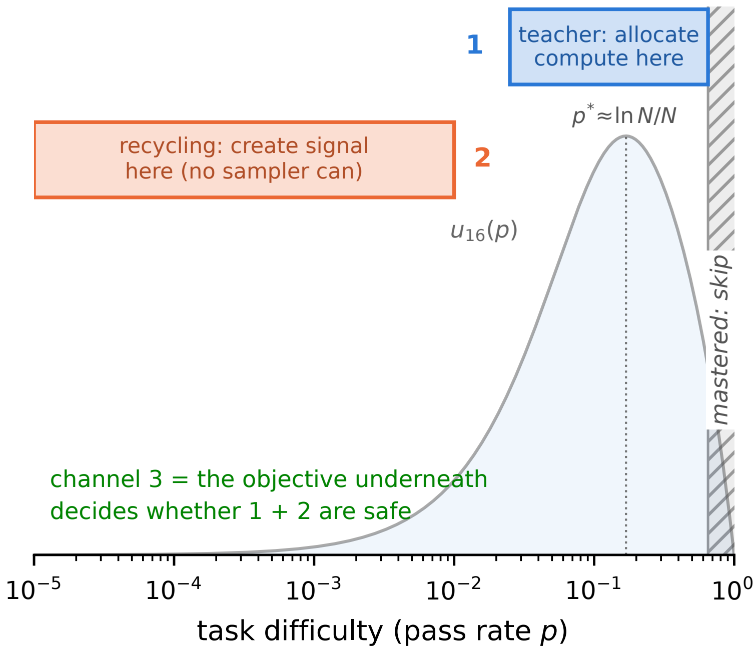

Figure 3: **Three channels.** The teacher allocates compute to the frontier band; recycling creates signal in the dead zone no sampler can reach; the objective underneath decides whether either is safe.

6 Three channels, one safety rule

The **teacher** avoids waste — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§7.1), so better pass-rate information is not where remaining gains live. **Recycling** creates signal — the only channel that still adds on top of the oracle. The **objective** underneath decides whether either is safe. One line: *the teacher allocates, hindsight creates, the objective decides whether either is safe.*

7 Experiments: an escalating ladder

7.1 Exact-gradient skill chains (CPU, 5 seeds). Uniform 0.650 → teacher 0.728 → full stack $\mathbf{0.890 \pm 0.002}$ AUC. The control battery locates where that comes from, and its eight arms order into a clean *gradient-information ladder*: zero-information dose (lr×2 0.798 ≈ random-target relabels 0.800) < reused correct direction (live-gradient replay 0.832) < new correct information (true relabels 0.869) < perfect allocation (oracle 0.8885 ≈ full stack 0.8895) < allocation+creation (oracle+recycling 0.8935) — each rung a 5/5 paired-seed win. *Allocation saturates:* the floor- and γ-matched true-pass-rate oracle ties the full stack; perfect information about pass rates is worth almost nothing over the cheap posterior. *Creation decomposes:* the replay placebo captures 83% of recycling's +0.22; the relabel direction carries +0.037 beyond it — and the direction term is the most *reliable* arm in the battery (seed spread ±.0025, 4–6× tighter than every placebo; its worst seed beats every placebo's best). Precision on the converse: random-target relabels are harmful *relative to dose-matched replay* (−.032, 5/5 seeds) but ≈ neutral vs. plain lr-doubling — fake labels waste the slot rather than poison training. Utility-form choice, same protocol: exact u_N 0.728±.002 ≈ legacy frontier form 0.733±.013 (the derived form's payoff is 7× smaller seed variance, not mean speed), but the N-blind N=2 learnability slice $p(1-p)$ forfeits the *entire* allocation gain — paired against uniform sampling it is a wash (−.007 mean, 2/5 wins) where u_N carries +.078 (5/5): the group size in the utility is load-bearing. Endpoint honesty: all arms finish within .027 of each other at convergence — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling (AUC spread 9× the final spread), which is exactly the currency — rollouts — that costs money at LLM scale.

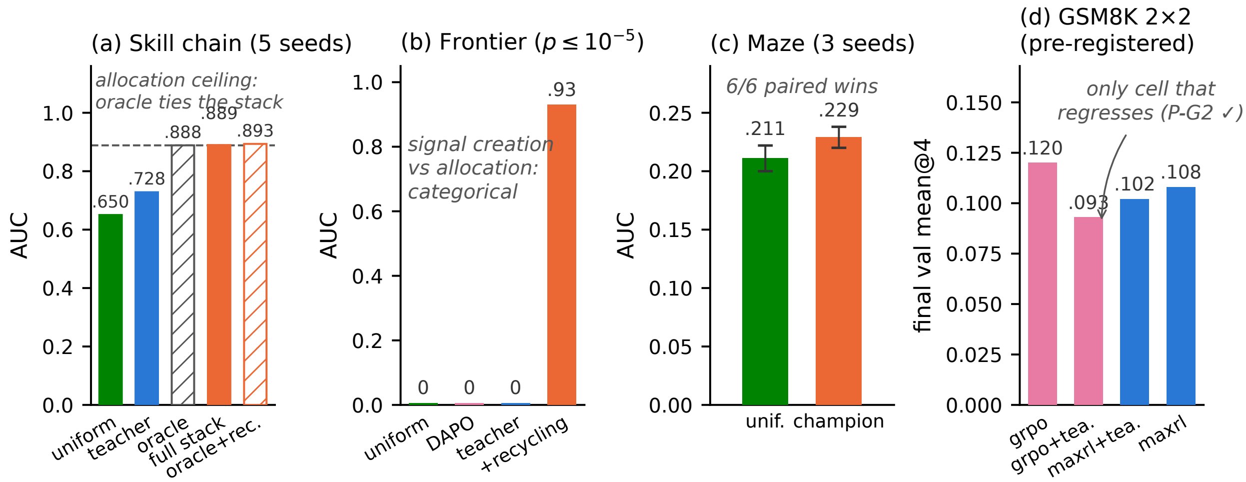

Figure 4: **The ladder.** (a) Exact-gradient chains: a floor-and-γ-matched true-pass-rate oracle *ties* the full stack — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; creation solves it. (c) Maze at matched wall-clock: 6/6 paired seed wins. (d) GSM8K 2×2, registered run: GRPO+teacher is the only cell that regresses (pre-registered P-G2; a replication seed did not reproduce it — §7.5); the teacher's MaxRL effect is a null at this budget (P-G1).

Takeaway. A monotone information ladder: dose < direction < allocation-ceiling < +creation. Allocation saturates; creation adds on top; and every intervention buys speed, not asymptote — efficiency is the honest currency.

7.2 Frontier-heavy regime (max pool pass rate 10^{-5}). Uniform, DAPO, and the plain teacher score **exactly 0.00**; teacher+recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ∼400 groups, then goes silent.

Takeaway. Where there is nothing to sample toward, only signal creation works — categorically, not marginally.

7.3 Maze transformer at matched wall-clock (1.26M params, 13 levels, ∼30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252±0.005 final / 0.229±0.009 AUC vs. uniform 0.230±0.015 / 0.211±0.011; paired deltas positive 6/6 seeds; 22–35% more optimization steps per GPU-hour. (Provenance note: the maze teacher predates the exact derivation and uses the legacy frontier form $(1-(1-p)^N)(1-p)$, plus an anti-forgetting term in the champion; the two forms' sampling distributions differ by total variation 0.013 at N=8, and a head-to-head on the CPU testbed shows them within noise — §7.1.) Safety, stated as what the design supports: an estimator *main effect on coverage* — all five MaxRL runs grow pass@8 (e.g. 0.316 → 0.348 under the teacher) and all four GRPO runs lose it (0.308 → 0.271 uniform; 0.332 → 0.269 with the teacher, the largest observed decay, though that arm is single-seed), permutation p=0.0079 on the 9-run main effect. Whether the teacher *amplifies* GRPO's decay is a factorial (interaction) question this suite is not powered for; the GSM8K 2×2 below tests exactly that cell, pre-registered. Inference currency: 1.2 × /2.7 × /**11**× fewer samples than GRPO to reach target coverage at levels 2/3/5 (honest 0.5× reversal at level 4); on the final uniform-teacher checkpoints the two estimators' summed coverage curves cross at k≈4 — GRPO ahead at pass@1 (3.24 vs. 3.07 summed over L0–6), MaxRL ahead from k=4 on and by half a level at k=64 (5.12 vs. 4.62; single checkpoint pair) — choose the estimator by how many samples you will draw at inference.

7.3b Where and when the coverage goes (level-resolved). Decomposing the main effect by difficulty band turns it into a mechanism (Fig. 5). *Where:* GRPO's pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 (−.224 mean over its four runs, every run negative) while its L1–3 pass@1 *rises*

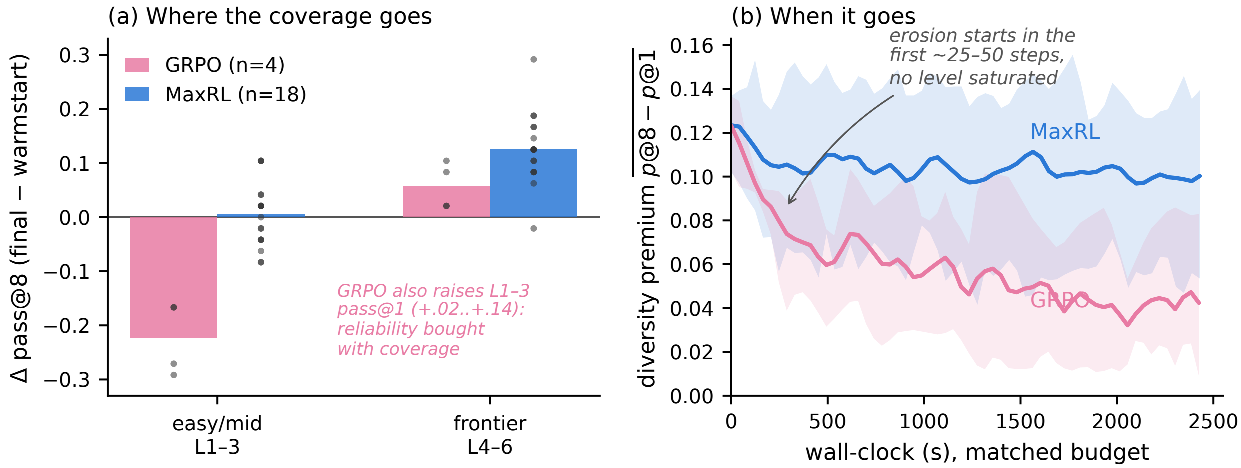

Figure 5: **Where and when GRPO's coverage goes (maze, level-resolved, matched wall-clock).** *(a)* Δpass@8 (final − warmstart) by difficulty band: GRPO's loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; band asymmetry exact permutation p=0.0001). *(b)* Diversity premium $\overline{\text{pass@8} - \text{pass@1}}$ vs. wall-clock (mean and min–max across runs): GRPO's erosion begins within the first ∼25–50 steps, before any level saturates; MaxRL stabilizes after warmup.

(+.02 to +.14, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat (+.005±.058 over 18 runs) and concentrates its gain on the frontier band L4–6 (+.126±.063 vs. GRPO's +.057). The band asymmetry is exact-permutation p=0.0001. This is the observable consequence of the mass curves in Fig. 1: the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts (sharpening them), where u_N has already released them. *When:* the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total diversity-premium loss ($\overline{\text{pass@8} - \text{pass@1}}$: .123 → .042) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only .62–.75; MaxRL's premium stabilizes after warmup (.122 → .101). A dose-response note on recycling from the same logs: sparse relabeling (1/dead group) is the only variant that preserves the diversity premium (Δgap −.005±.009 vs. −.030 for both no-recycling and dense; exact p=0.007/0.008), while dense banks the same coverage as sparse into the best final pass@1 — on the maze, dose sets where the coverage dividend is spent, not whether coverage is lost (final pass@8 across doses: null, p=0.39).

Takeaway. The gains survive real gradients, and the coverage main effect now has a located mechanism: GRPO liquidates easy-band coverage for reliability from the first updates; MaxRL reallocates the same budget to the frontier. Coverage is the currency that sees it.

7.4a Legged locomotion (IsaacLab, pre-registered honest null). A 5-arm pilot on Anymal-C rough terrain (frontier teacher vs. stock greedy walker, uniform, scripted ramp, and a no-motion control; fixed-grid per-level evaluation, one seed, run by an independent team on a co-designed protocol) lands the *pre-registered* null exactly: on a terrain grid where every level is learnable at budget, the stock greedy curriculum wins outright (mean pass .372 vs. the teacher's .278), and the teacher's only edge is a within-noise sliver at the hardest evaluated level. Combined with the maze step-matched analysis and Countdown, this makes the regime rule a three-scale result: **allocation pays only where unlearnable-at-budget regions exist to avoid** — on learnable-everywhere pools, breadth beats focus at 1.26M-param maze, 360M LLM, and legged-robot scale.

7.4b Gymnasium control (MountainCar, CartPole) — binary-success RLVR on classic control, official-task currency. Setting, stated precisely because the baseline label matters: all arms train on *binary task success* (MountainCar: reach $x \geq x^*$; CartPole: survive $\geq T$) with the same group estimator —

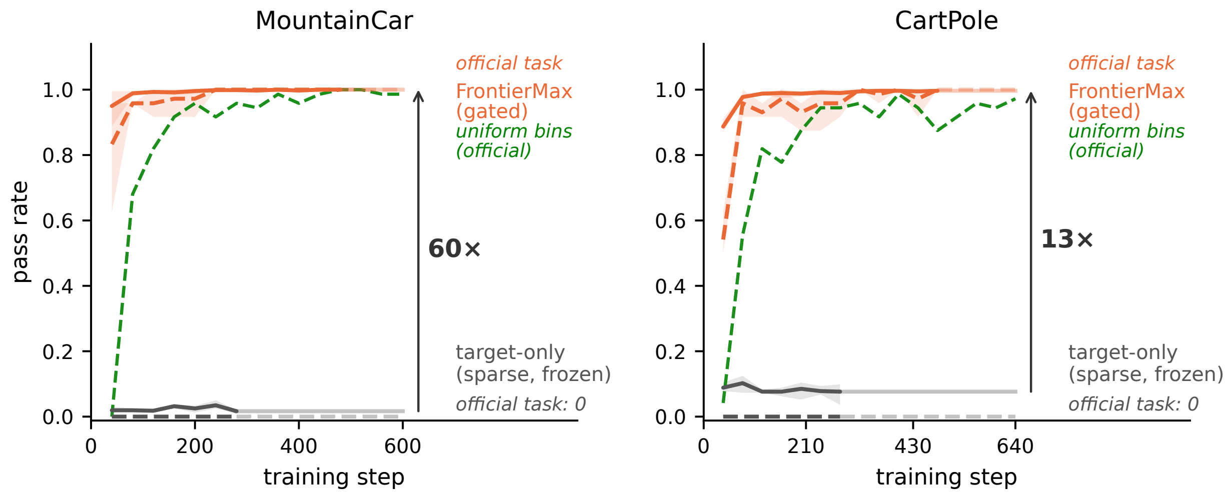

Figure 6: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward — all-fail groups emit zero gradient in every group-relative estimator (with CartPole's native dense reward, plain REINFORCE solves the task; the zero is a property of the sparse setting this rung models). A uniform-over-bins control (green) also cracks the task — 3× slower to hard-task 0.9 and off-ceiling — isolating the teacher's contribution (speed, seed-stability) from task spread's (existence of the gradient); the gated FrontierMax stack converges to 1.000±0.000.

the classic-control analogue of a verifier reward, not the environments' native dense rewards that standard libraries train on. The **target-only** arm (sample only the official task) is then a provable zero-update freeze, not a slow learner: every group is all-fail, every group-relative estimator (MaxRL, GRPO, RLOO) emits exactly zero weights on constant rewards, and telemetry confirms the policy received *zero* parameter updates over its full budget ($\|\theta\| = 0$ throughout; measured random-policy success is $< 3{\times}10^{-6}$ for the flag over 10^6 episodes and $\approx 10^{-13}$ for survive-400 by tail fit — no budget ignites it). The gated full stack under the identical reward reaches **1.000±0.000 on both official tasks** (3 seeds, Fig. 6). Honesty about the dense-reward world: with CartPole's native +1/step reward, vanilla REINFORCE on the *same tile policy* solves survive-400 within this budget (0.50–0.75 by 4,000 episodes) — the categorical zero is a property of sparse verifier-style reward, which is exactly the RLVR setting this rung models; MountainCar's native reward (−1/step, constant until first success) flatlines policy gradients either way, which is why it is *the* hard-exploration classic. Attribution within the sparse setting, three rungs at a common 280-step horizon (hard-task AUC): uniform-over-bins task spread gets most of the way (MC .750 → +recycling .895 → +teacher .956; CartPole .708 → .792 → .893; the paired teacher increment is positive in 3/3 seeds on both envs), creation-without-allocation plateaus *below* ceiling on CartPole (0.958 in every seed) while adding the teacher restores 1.000, and the teacher's remaining contribution is speed and stability: 3.0× faster to hard-task 0.9 on both environments (2.1× to 0.99 on CartPole) with 2.5–6× smaller cross-seed variance along the whole curve. An instructive methods note: our own first version of this study silently used per-bin task-partitioned policy parameters and reproduced the known failure (flag 0.000 in every arm) — restoring the validated shared-policy design (the task enters only the success predicate) is what cracks the task, replicating the 10-seed transition-matched branch study (flag 0.848±0.058) and strengthening it to full convergence.

Takeaway. Curricula operate through shared parameters, or not at all.

7.5 LLM scale: GSM8K 2×2, pre-registered (SmolLM2-360M, N=16, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run:** GRPO+teacher was the only cell that regressed in its second half (val accuracy .096 → .093, pass@4 .193 → .181) while plain GRPO climbed monotonically (.078 → .105 → .120) — with the teacher verifiably steering (sampled-dead fraction driven to

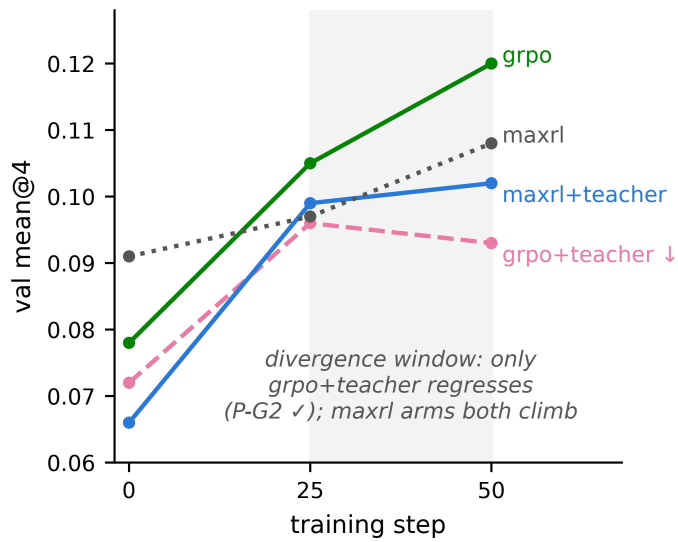

Figure 7: **GSM8K 2×2 (pre-registered; registered run shown).** GRPO+teacher is the only cell that regresses in its second half (P-G2 on this run; a replication seed with weaker measured steering climbed instead — §7.5); under MaxRL the identical teacher trains stably to its best value but does not beat uniform at this budget (P-G1: null).

0.48 vs. the ∼0.65 population rate). **A replication seed does not reproduce the regression:** seed 2 of the same cell climbs monotonically (.073 → .095 → .118), and its steering telemetry shows the treatment was weaker (min dead-sampled fraction .53, mean .66 ≈ population — a near-uniform teacher on average). The pattern is mechanistically consistent — damage tracks steering intensity, and seed 2's final pass@4 (.203) still sits below uniform GRPO's (.229) — but the honest status of the LLM-scale interaction is *1 of 2 seeds, treatment-intensity-dependent*: suggestive, not established. MaxRL+teacher trains stably to its best value (.066 → .102); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) an honest *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@k sweep on the registered checkpoints: the teacher's deficit under GRPO *widens monotonically with* k (−.008 at k=1 to −.024 at k=16) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise. Beyond the predictions: the teacher's posterior *learns* real difficulty from ∼1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$) but its allocation is *posterior-starved* — 3,200 draws over 7,473 prompts leave Thompson sampling near-uniform, and seed 2 shows that starvation can neutralize the treatment itself. Two posterior diagnostics sharpen this. *The map is real:* two independent teacher runs' posteriors agree with each other (cross-run Spearman ≈.45 on co-visited prompts, n≈900, ≈ .67 self-consistency with a shared latent) three times better than either agrees with external 7B-model difficulty annotations (.10–.22) — the teacher learns *this model's* difficulty landscape, reliably, from ∼1.25 visits per prompt; the published $\rho = -0.17$ against the 7B key understates what the posterior knows. *But it is compressed:* across external-difficulty quintiles spanning solve rates .33 → .98, mean $\hat{p}$ moves only .076 → .103 — Thompson sampling sees a nearly flat utility landscape, which *is* the allocation-starvation mechanism. A sharpening note in a different currency: response length separates the runs by estimator, perfectly — all three GRPO runs compress to 209.6±1.8 tokens while both MaxRL runs sit at 232.4±5.7 (common ∼365-token warmstart; the teacher moves length ≤ 3 tokens): the estimator, not the curriculum, sets how hard the policy templates its outputs, visible even where token entropy is ambiguous. Small model; the pre-registered outcomes were orderings and signs on the registered run.

Takeaway. The maze main effect stands on nine runs; the LLM-scale interaction is 1-of-2-seeds and tracks steering intensity — the decisive experiment is a steering-controlled multi-seed cell, queued.

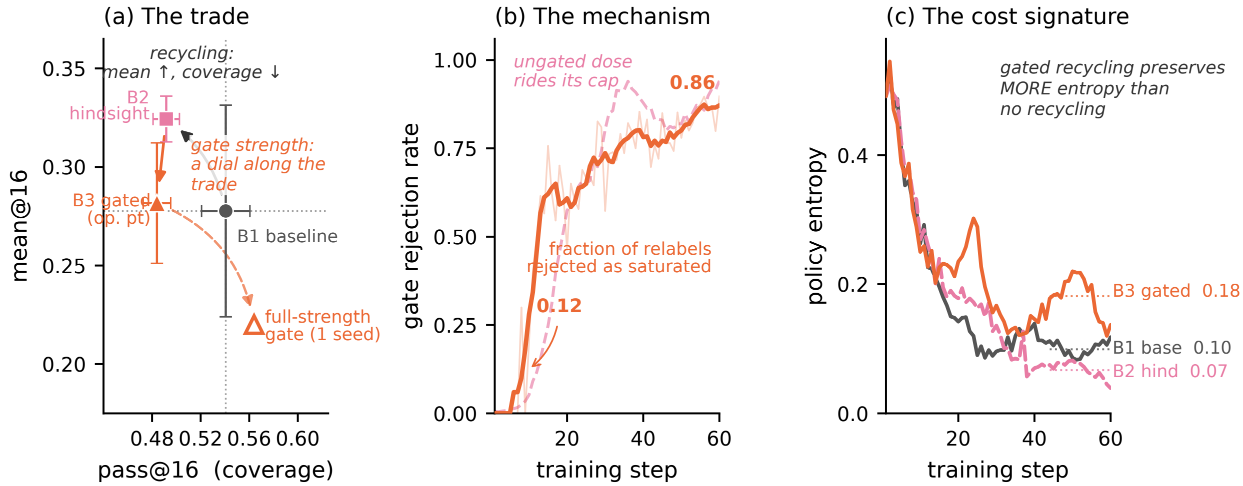

Figure 8: **Recycling-induced sharpening and its derived mitigation (Countdown, tier 1).** *(a)* Three seeds, both axes: recycling (B2) buys mean@16 with pass@16 coverage. On this tier the moderate gate (B3) gives back most of the mean without moving coverage — tier-1 destinations saturate early, so admitted relabels are few — while full-strength gating (corrected decay, 1 seed) returns coverage *above* baseline at the mean's full price: gate strength is a dial along the trade, and its useful middle lives on the frontier tier, where the same moderate setting restores coverage to baseline (.279±.019 vs. .274) keeping ~60% of the mean gain (§7.6). *(b)* The mechanism in-run: the gate's rejection rate rises monotonically as the model's reachable set saturates its relabel destinations, while the ungated dose rides its cap. *(c)* Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

7.6 Countdown 2×2 (pre-registered): two honest nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool's simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §7.3's weight-induced collapse — data-induced rather than weight-induced, and to our knowledge unreported (no prior work measures pass@k under hindsight in RLVR). Our algebra predicts it: a relabel to a self-achieved value is a task at $p \approx 1$, where $u(p)$ says training signal is zero — the softmax pays for those updates out of the exploration tail. *Takeaway: recycling needs an admission rule, and the estimator already provides it — the same derived utility that schedules sampling should gate the relabel destination.*

7.7 The replication and the gate (Countdown v2 pool, 3 seeds, pre-registered). We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms × three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). Setup note with a theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures (relabel yield 0.000, n=384), so there is nothing to recycle until SFT installs the output format; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant ~75% of the independence bound $1-(1-\text{pass@1})^{16}$ (ratios .744/.783/.755 across tiers) — the baseline our headroom gate measures against. **Sharpening replicates:** at tier 1, recycling buys mean@16 +.046 (.278±.054 → .324±.012) and pays −.049 of pass@16 (.541±.020 → .492±.011), both outside seed noise; the frontier tier moves the same direction. **The timing is dose-then-damage:** recycling's frontier-tier mean gain is installed by step 30 — +.014±.008 paired, all seeds positive, at only ~28% of the eventual relabel dose — while the coverage cost only materializes by step 45 (−.048±.019 paired, *all* *seeds negative*, after ∼65% of dose): the benefit precedes the damage, so early-stopping on mean-accuracy validation would bank the gain and never see the bill. The damage is also visible on the training side: the ungated arm ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired $+.083/+.063/+.034$). And the 3-seed entropy read sharpens the single-seed one: the *gate's* entropy retention replicates (+.072 paired over baseline, all seeds positive, ∼1.6×), while ungated recycling makes entropy *seed-unstable* (endpoint spread ±.062, 4–9× the other arms') — consistent with its noisy frontier-coverage endpoints. **The gate works where the algebra says it should:** at the frontier tier (destinations not yet saturated) the gated arm returns coverage to baseline (.279±.019 vs. .274±.015) while keeping ∼60% of the mean gain; at tier 1 (destinations saturate early, so the gate blocks nearly all relabels) it neutralizes recycling almost entirely, and a corrected-decay full-strength rerun (1 seed) pushes tier-1 coverage above baseline at the mean's full price (Fig. 8a). In-run telemetry shows the mechanism (Fig. 8b–c): rejection rises .12 → .85 as the reachable set saturates, and the gated arm ends with *more* policy entropy than the no-recycling baseline. A pre-registered `gate_max_p` dose sweep at fixed decay is the standing follow-up.

Takeaway. Sharpening is a reproducible property of exact-verifier recycling, and the derived gate converts it into a tunable mean-vs-coverage trade whose useful range sits where destinations are not yet saturated.

8 Related work

The coverage tension. RLVR itself trades coverage for precision: pass@1 rises while pass@k at large k falls below the base model [4], with entropy collapse as the mechanistic signature [6] and overtraining on saturated problems as a locus [5]. Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator's implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss [7]; our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1-\hat{p})$, which Prop. 1's corollary identifies as the $N{=}2$ slice of our derived utility. Reinforce-Ada [13] — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and Prop. 3 is the idealized form of both (water-filling spends its next rollout wherever $p(1-p)^N$ is largest, across or within). DAPO's dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward pass@k but keep uniform sampling. Hindsight for LLMs (AgentHER, HSL) relabels agent trajectories with an LLM judge. The nearest neighbor is concurrent: LfH [18] brings hindsight relabeling into GRPO for VLA post-training — a VLM proposes one shared hindsight instruction per failed group and rescores the group under it; 5× sample efficiency on LIBERO-PRO. LfH's relabeler is a VLM judge (their stated limitation: relabel noise), and its evaluation never measures pass@k — precisely the currency where we find recycling's hidden cost. Exact-verifier relabeling with a correctness guarantee, and the coverage accounting of recycling, remain, to our knowledge, ours. No prior work (i) derives the sampling utility from the estimator's own expected signal, (ii) couples it with success-conditioned weights it provably matches, and (iii) adds a signal-creation channel with an exactness guarantee.

9 Limitations and honest negatives

Deep frontiers at fixed budget remain uncrossed on the maze, and a 4× duration run *refuted* our own duration hypothesis (the stall is a per-step-competence ceiling; depth needs capacity, not more schedule). γ-concentration did not transfer from chains to the maze — predicted in advance by the compounding ODE model. Adaptive truncation order underperformed fixed T. Feeding relabeled successes to the teacher's posterior inflates it — dropped. An early GSM8K claim of teacher steering was retracted when review found the sampler epoch-frozen; the fixed run confirmed the effect properly; an early "beats the oracle" claim was retracted when a floor-handicap was found in the oracle arm (§7.1 reports the corrected tie). Recycled-gradient exactness is a property of the environment's relabel map; noisier verifiers will land between our +0.22 (fixed pools) and +0.01 (one-shot) endpoints. Countdown results are 3-seed. On GSM8K the replication seed of the critical GRPO+teacher cell did *not* reproduce the registered regression, with measurably weaker steering (§7.5) — so the LLM-scale teacher×estimator interaction is 1-of-2 seeds and treatment-intensity-dependent, and we do not claim it as established; the GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there is about the sign of the mean-accuracy effect, not a coverage rescue. The maze estimator main effect (nine runs, zero exceptions) is the safety result this paper stands on; a steering-controlled multi-seed LLM cell is the decisive next experiment.

Reproducibility

Every proposition has a Monte-Carlo verification script; every figure in this paper regenerates from JSON artifacts committed to the repository (`paper/figures/`, with LLM-run endpoints vendored in `paper/figures/data/`); the documentation passed a two-round adversarial audit. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team's summary (raw logs live with that team), and per-step timing fractions come from verl's in-run telemetry in the ray session logs. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.

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